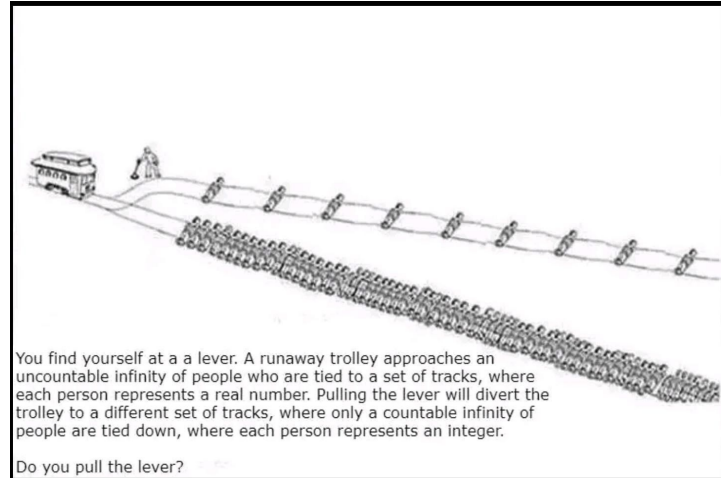


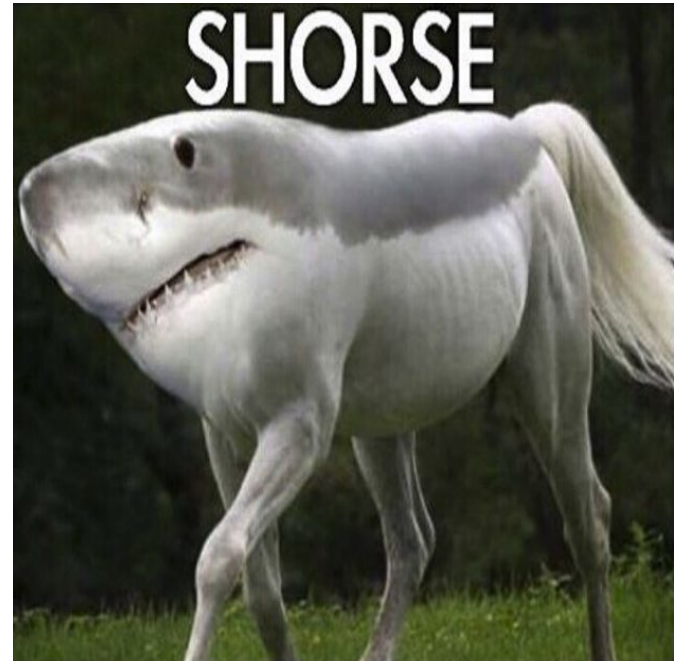
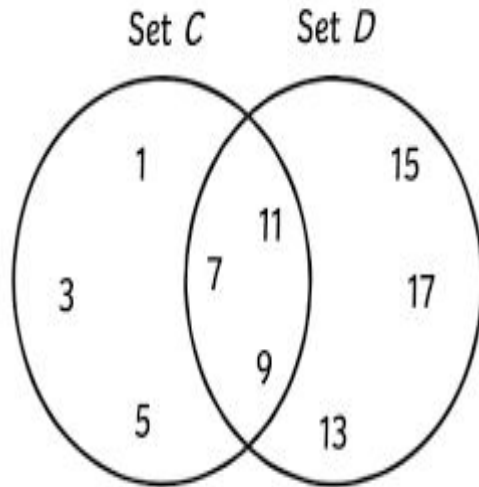
Countable vs Uncountable Infinity

Click to add subtitle



What is a set?

A set is a collection of distinct objects, called elements or members, considered as a single mathematical entity.



What is counting?



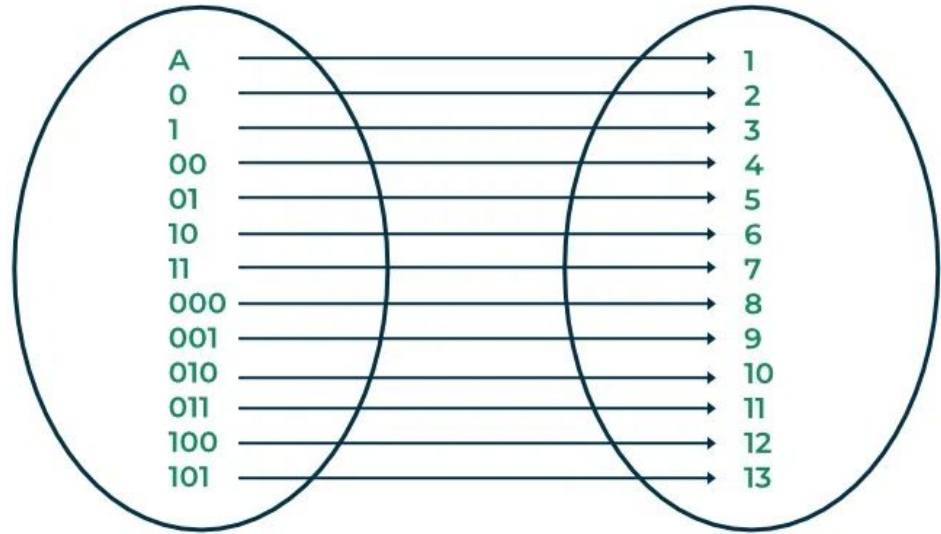
Countable sets?

Natural numbers $\mathbb{N} = \{1, 2, 3, 4, \dots\}$

Even numbers $\{2, 4, 6, 8, \dots\}$

Perfect squares $\{1, 4, 9, 16, 25, \dots\}$

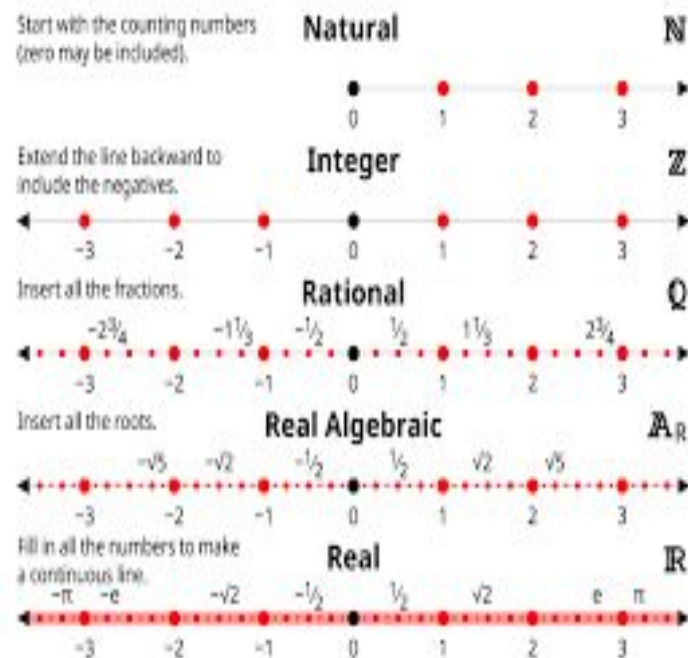
Integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$



Uncountable sets?

Real numbers $\mathbb{R}$

Real numbers between 0 and 1: $[0, 1]$



Cantor Set

Start with the interval $[0, 1]$

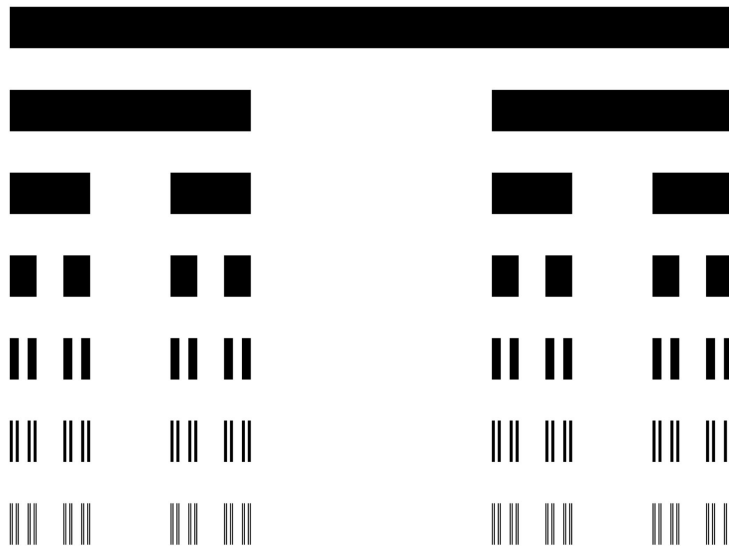
Step 1: Remove the middle third $(1/3, 2/3)$

- Left with: $[0, 1/3] \cup [2/3, 1]$

Step 2: Remove the middle third of each remaining piece

- Remove $(1/9, 2/9)$ and $(7/9, 8/9)$
- Left with: $[0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1]$

Step 3: Keep removing middle thirds forever...



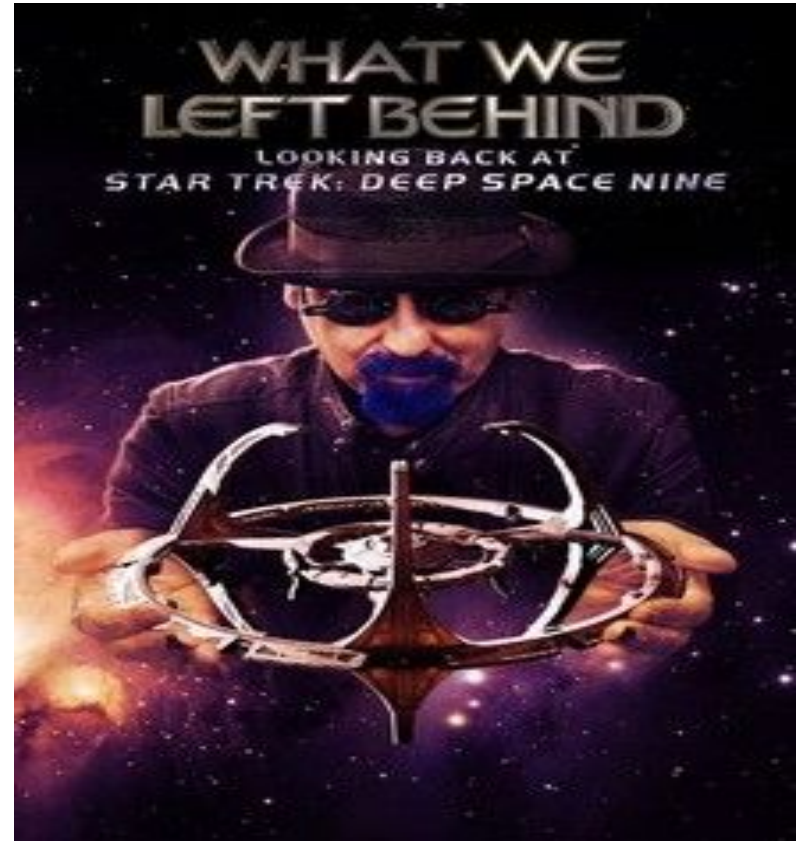
Whats left

What's Left?

The Cantor Set = all the points that never get removed

Points that survive:

- All the endpoints: 0, $\frac{1}{3}$, $\frac{2}{3}$, 1, $\frac{1}{9}$, $\frac{2}{9}$, $\frac{7}{9}$, $\frac{8}{9}$, ...



But

Freaky points left also left

Non-endpoint points are where sequences of endpoints cluster

Example: The sequence $1/3, 7/27, 25/81, 79/243, \dots$ approaches a limit point

That limit isn't necessarily an endpoint, but it's in the Cantor Set

